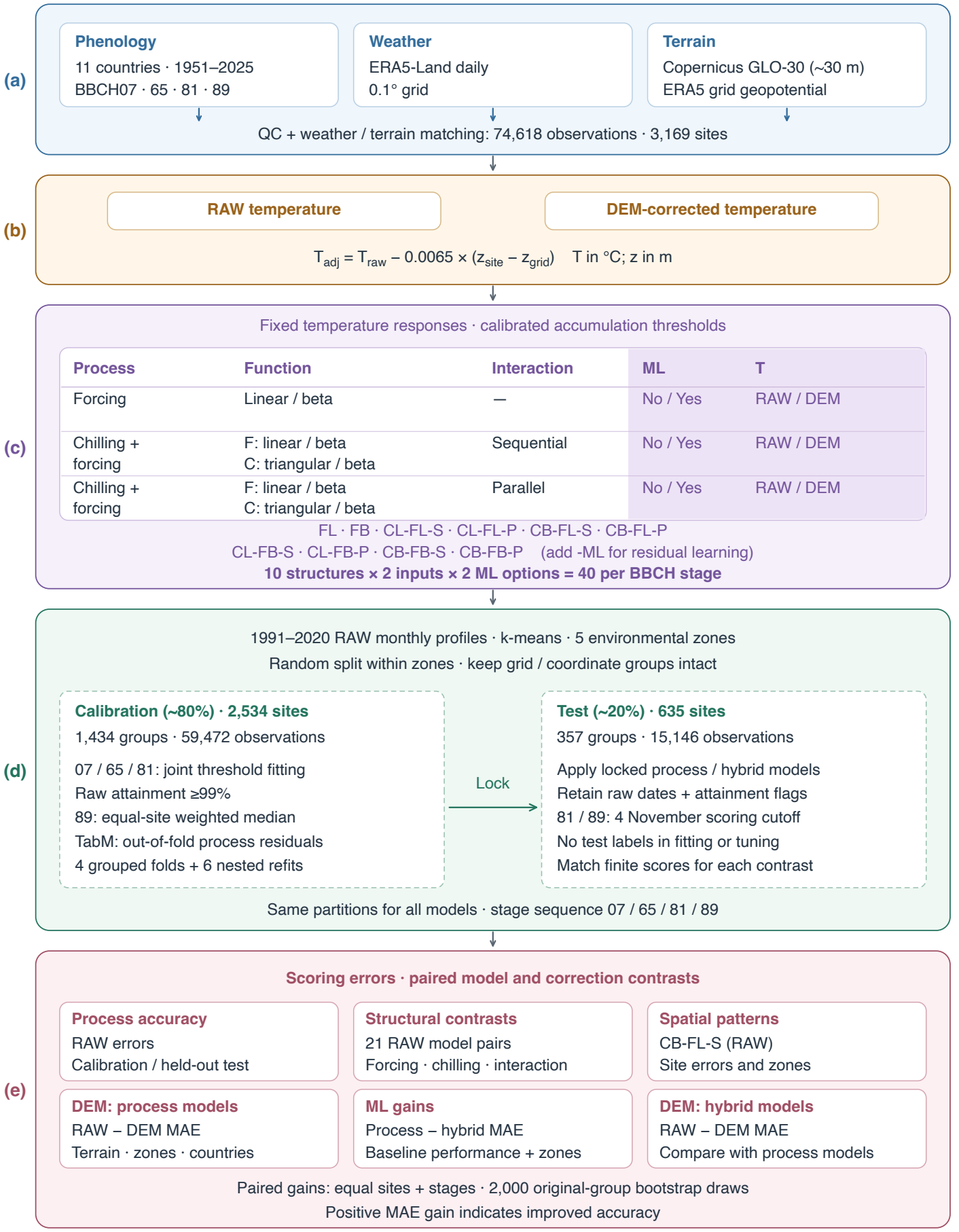

(a)
Phenology
11 countries · 1951–2025
BBCH07 · 65 · 81 · 89
Weather
ERA5-Land daily
0.1° grid
Terrain
Copernicus GLO-30 (~30 m)
ERA5 grid geopotential
QC + weather / terrain matching: 74,618 observations · 3,169 sites
(b)
RAW temperature
DEM-corrected temperature
$T_{adj} = T_{raw} - 0.0065 \times (z_{site} - z_{grid})$ T in °C; z in m
(c)
Fixed temperature responses · calibrated accumulation thresholds
Process | Function | Interaction | ML | T
Forcing | Linear / beta | — | No / Yes | RAW / DEM
Chilling + forcing | F: linear / beta C: triangular / beta | Sequential | No / Yes | RAW / DEM
Chilling + forcing | F: linear / beta C: triangular / beta | Parallel | No / Yes | RAW / DEM
FL · FB · CL-FL-S · CL-FL-P · CB-FL-S · CB-FL-P
CL-FB-S · CL-FB-P · CB-FB-S · CB-FB-P (add -ML for residual learning)
10 structures × 2 inputs × 2 ML options = 40 per BBCH stage
(d)
1991–2020 RAW monthly profiles · k-means · 5 environmental zones
Random split within zones · keep grid / coordinate groups intact
Calibration (~80%) · 2,534 sites
1,434 groups · 59,472 observations
07 / 65 / 81: joint threshold fitting
Raw attainment ≥99%
89: equal-site weighted median
TabM: out-of-fold process residuals
4 grouped folds + 6 nested refits
Lock
Test (~20%) · 635 sites
357 groups · 15,146 observations
Apply locked process / hybrid models
Retain raw dates + attainment flags
81 / 89: 4 November scoring cutoff
No test labels in fitting or tuning
Match finite scores for each contrast
Same partitions for all models · stage sequence 07 / 65 / 81 / 89
(e)
Scoring errors · paired model and correction contrasts
Process accuracy
RAW errors
Calibration / held-out test
Structural contrasts
21 RAW model pairs
Forcing · chilling · interaction
Spatial patterns
CB-FL-S (RAW)
Site errors and zones
DEM: process models
RAW – DEM MAE
Terrain · zones · countries
ML gains
Process – hybrid MAE
Baseline performance + zones
DEM: hybrid models
RAW – DEM MAE
Compare with process models
Paired gains: equal sites + stages · 2,000 original-group bootstrap draws
Positive MAE gain indicates improved accuracy